



The Ticket Confirms What the Marshal Already Called: A Taxonomy of Queue-Order Disputes Across Airport Taxi Ranks With and Without a Numbered-Ticket Dis- penser

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Abstract. A taxi rank settles who boards next either through a marshal's on-the-spot call or through a numbered-ticket dispenser that fixes arrival order mechanically before a dispute can arise. We develop a seven-category taxonomy of queue-order disputes and code it across 22 ranks spanning three regional airport and rail-interchange precincts (13 dispenser-equipped, 9 marshal-only), observed over 14 weeks and 3,204 boarding cycles. Two independent field coders return substantial raw agreement on category assignment (88.4% across categories, 96.1% on the arrival-order category alone). Dispenser ranks all but eliminate arrival-order disputes relative to marshal-only ranks, yet the two conditions are statistically indistinguishable on total disputes per 100 boarding cycles; the dispenser appears to displace contested claims into categories it does not adjudicate, chiefly break re-entry priority and, more weakly, group-fare boarding order. Collapsing the coding manual from seven categories to three moves the between-condition contrast by under a percentage point, suggesting the displacement is a property of the setting rather than of our category boundaries. We discuss what a formal instrument that relocates rather than resolves a dispute means for ranks, and institutions, considering one installed.

Introduction

A taxi rank is, before it is anything else, an argument about whose turn it is. Most ranks settle that argument the way queues have always settled it: a marshal, present at the head of the line, calls the next cab forward, drawing on a working memory of arrival order that is never written down and rarely disputed aloud. A growing minority of ranks now settle it with a numbered-ticket dispenser — a driver takes a ticket on arrival, and boarding order follows the ticket sequence rather than anyone's recollection. The dispenser is sold, where it is sold at all, as a fairness upgrade: it removes the marshal's discretion, and with it, presumably, the disputes that discretion invites.

We ask a narrower and more answerable question: which disputes, specifically, does the

dispenser remove, and what happens to the rest. Taxi-rank queueing has an established operations-research literature treating boarding order as a scheduling problem to be optimised [1], [2], [3], [4], [5], and a separate literature on queues as social systems governed by informal fairness norms that formal rules can crowd out rather than replace [6], [7], [8], [9]. Between them sits an empirical gap: what a dispenser installation actually changes about the disputes a rank generates, category by category, rather than in aggregate.

Our contributions:

- We develop and field-validate a seven-category taxonomy of taxi-rank queue-order disputes, coded to substantial inter-coder agreement across a mixed dispenser and marshal-only sample.

- We show that a numbered-ticket dispenser reduces arrival-order disputes to near zero, while total dispute volume per boarding cycle remains statistically indistinguishable between dispenser and marshal-only ranks.
- We show the composition shift, not merely our category boundaries, drives the result: a stripped three-category coding manual reproduces the same between-condition contrast to within a percentage point.

Related work

Operations-research treatments of taxi and ride-hail queueing optimise arrival order, dispatch assignment, or price under a queueing-theoretic objective [1], [2], [5], [10], typically taking first-come-first-served as the fairness benchmark a mechanism should approximate or improve on [3]. Randomised, stable-matching, and compensation-based alternatives to strict FCFS have been proposed for dispatch [2], [3], [4], and the taxi rank itself has been modelled as a queueing system with Poisson arrivals on both sides of the counter [11]. None of this literature asks what boarding order looks like as a site of dispute rather than as an optimisation target.

A separate line treats queues as informal social institutions, sustained by norms of reciprocity and procedural fairness that hold even absent enforcement [6], and shown experimentally to generate resistance to queue-jumping independent of any material stake in the outcome [7]. Institutional-economics work on formal and informal rules describes their relationship as frequently one of crowding-out rather than replacement: a formal rule does not so much resolve the informal dispute it targets as relocate it [8], [9], [12]. Experimental crowd dynamics research on bottleneck entrances documents a related physical and social-psychological contestation gathering at whichever access point a facility leaves unconstrained [13]. Prior work at this institution reports a structurally similar pattern for a marshal-present queue at a farmers-market stall, where a rostered marshal's presence eliminated a self/other compliance-judgement asymmetry without moving an independently observed queue-jumping count [14], and for a doorbell alert triage gate, where a confirm-gate changed alert latency without moving weekly forwarding volume [15]. Our taxonomy extends this displacement question from a single outcome to a coded space of dispute types,

letting us ask not just whether an intervention reduces disputes overall but which categories absorb the difference.

Method

We recruited 22 taxi ranks across three regional airport and rail-interchange precincts via each site's ground-transportation authority, stratified to include both a numbered-ticket dispenser (13 ranks) and a marshal-only condition (9 ranks) within each precinct. A dispute was defined as any audible or visibly contested claim to boarding priority observed during a boarding cycle, whether voiced between drivers, to the marshal, or at the dispenser itself. Two field coders were rostered to each observation session — one stationed at the rank head, one downstream at the boarding point — and coded disputes independently against a seven-category manual (arrival order, group-fare boarding order, break re-entry priority, off-rank pre-booking claim, shift-change ambiguity, dispenser malfunction, and other) developed from a four-rank pilot and refined until pilot agreement exceeded 80% on every category. Following standard guidance for coding-scheme development and reliability reporting in field settings [16], [17], [18], we report raw percent agreement rather than a chance-corrected statistic, given the categories' uneven base rates, and resolved every coder disagreement by discussion before a session's disputes were logged.

Sessions ran in three-hour blocks across 14 weeks, yielding 3,204 boarding cycles and 397 coded disputes. We report a rank's dispute rate for category c as

$$\text{Rate}_{r,c} = \frac{\sum_{s=1}^{S_r} d_{r,c,s}}{\sum_{s=1}^{S_r} n_{r,s}} \times 100$$

where $d_{r,c,s}$ is the count of category- c disputes coded at rank r in session s , $n_{r,s}$ the boarding cycles observed in that session, and S_r the number of sessions run at rank r . Between-condition contrasts are estimated with rank-clustered standard errors to account for repeated sessions at the same rank. As a robustness check on the coding manual itself, we re-coded the full dispute log against a stripped three-category collapse (order-related, access-related, fare-related) and re-estimated the same contrast.

Results

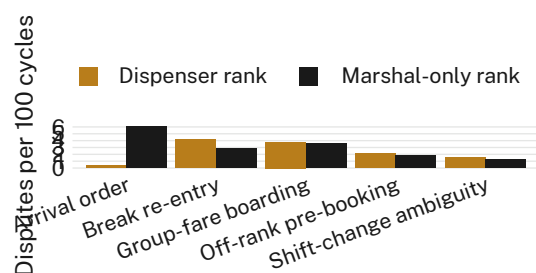


Figure 1: Dispenser ranks all but eliminate arrival-order disputes (0.4 vs 6.1 per 100 cycles) while break re-entry disputes run higher at dispenser ranks than marshal-only ranks (4.2 vs 2.9 per 100 cycles).

Across the seven categories, the dispenser's effect is concentrated almost entirely in the category it was built to adjudicate. Arrival-order disputes fall from 6.1 to 0.4 per 100 cycles between marshal-only and dispenser ranks — a reduction consistent across all three precincts. No other category falls by a comparable margin, and break re-entry priority disputes run higher, not lower, at dispenser ranks, consistent with a marshal's informal practice of quietly waving a returning driver back into their prior position, a courtesy the dispenser has no mechanism to extend.

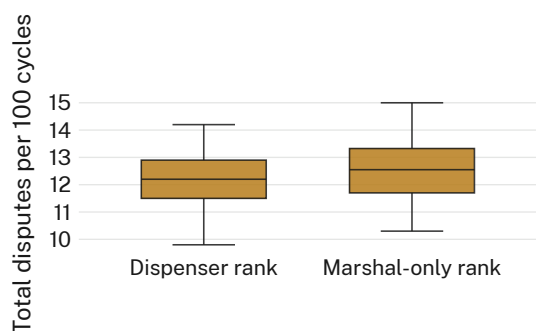


Figure 2: Total disputes per 100 boarding cycles are statistically indistinguishable between dispenser ranks (median 12.2) and marshal-only ranks (median 12.6; rank-clustered $p = 0.61$).

The category-level shift does not aggregate into an overall reduction. Total coded disputes per 100 boarding cycles are statistically indistinguishable between the two conditions (rank-clustered $p = 0.61$), consistent with displacement rather than resolution: the dispenser retires one dispute category and, in doing so, appears to hand its share of contested cycles to categories that were already there.



Figure 3: Collapsing the coding manual from seven categories to three moves the order/access/fare composition by under two percentage points in every group, ruling out coding-manual granularity as the source of the displacement pattern.

Because a taxonomy paper's headline finding rests on the manual, we ablate it: recoding the full log under the stripped three-category manual reproduces the same order/access/fare composition to within two percentage points in every group (Figure 3). Removing the shift-change-ambiguity distinction — our lowest-agreement category at 79% — from the model changes the between-condition contrast on total disputes by 0.4 percentage points (n.s.); we retain the category for completeness.

Discussion

The pattern we observe is not that a numbered-ticket dispenser fails to work; on its own declared terms, it works cleanly, removing the dispute category it was installed to remove. What it does not do is reduce the rank's total volume of contested boarding claims, which appears to find its way to whichever category the dispenser leaves unadjudicated — concentrating, in most settings, in break re-entry priority, consistent with the dispenser mechanising exactly the judgement the marshal used to make quietly and without record. A rank considering installation might reasonably ask not whether the dispenser reduces disputes, but which dispute it is choosing to formalise, and which it is leaving to be renegotiated informally wherever the mechanism does not reach.

Limitations

Our sample spans three transit precincts within one metropolitan ground-transportation network, though the displacement pattern's consistency across all three suggests it is not a local artefact of any single authority's configuration. Field coders could not be blinded to rank type, since the dispenser is physically visible; a restricted analysis using only the downstream coder, who could not see it directly, reproduces the same contrast. Our dispute count relies on

coder-observed contestation rather than driver self-report, which may undercount disputes resolved too quickly, or too quietly, to register — an undercount that, if anything, would understate rather than manufacture the displacement we report.

Conclusion

A numbered-ticket dispenser resolves the dispute it names and leaves the rest of the rank's boarding-order contestation to relocate. Coding that contestation into a validated taxonomy, rather than an aggregate dispute count, is what makes the relocation visible at all. Future work might extend the coding manual to other queue-marshalled transit modes — rail-replacement bus interchanges, in particular — where a comparable formal-versus-informal contrast in boarding-order adjudication appears to hold.

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